# IGCSE 0607 Practice Questions — E5.2

## Angle measurement in degrees

**Total Questions:** 6 — **Total Marks:** 40 Marks

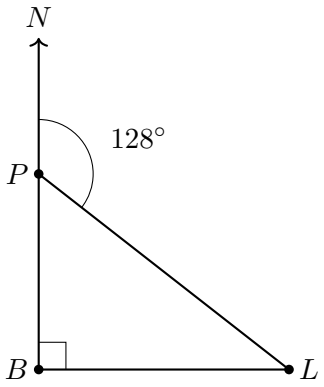
---

## Table of Contents & Quick Navigation


- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*


---

**1** The bearing of a lighthouse $L$ from a port $P$ is 128°.
A boat $B$ is positioned such that it is due South of $P$ and due West of $L$.




**(a)** Find the bearing of $P$ from $L$.

..... ° **[2]**

**(b)** Work out the size of angle $BPL$.

..... ° **[1]**

**(c)** Work out the size of angle $PLB$.

..... ° **[2]**

**2** The bearing of a lighthouse, $L$, from a boat, $B$, is 148°.

**(a)** Work out the bearing of $B$ from $L$.

..... ° **[2]**

**(b)** A port, $P$, is due north of $L$ and due east of $B$.

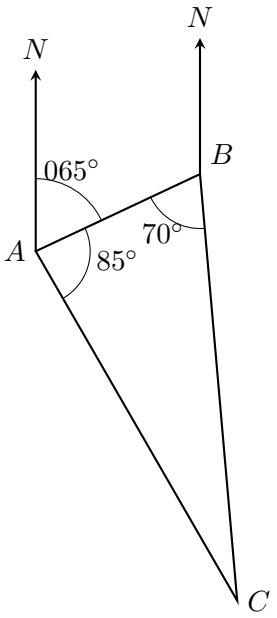
**(i)** Write down the bearing of $B$ from $P$.

..... ° **[1]**

**(ii)** Work out the size of angle $LBP$.

..... ° **[2]**

**3** The diagram shows the positions of three towns, $A$, $B$ and $C$.
The bearing of $B$ from $A$ is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.




**(a)** Calculate the bearing of $C$ from $A$.

..... ° **[1]**

**(b)** Calculate the bearing of $A$ from $B$.

..... ° **[2]**

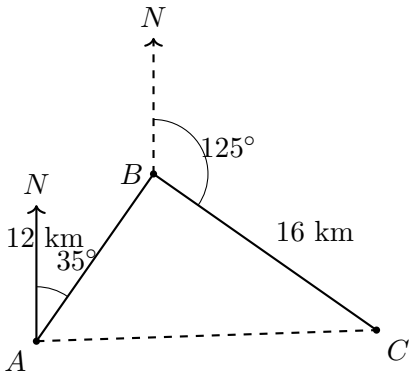
**(c)** Calculate the bearing of $C$ from $B$.

..... ° **[2]**

**(d)** Calculate angle $ACB$.

..... ° **[1]**

**4** A ship leaves a port $A$ and travels 12 km on a bearing of 035° to point $B$.
From $B$, the ship travels 16 km on a bearing of 125° to point $C$.




**(a)** Show that angle $ABC = 90°$.

**[2]**

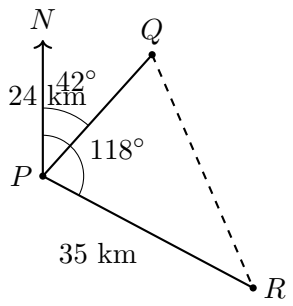
**(b)** Calculate the direct distance $AC$.

$AC =$ ..... km **[3]**

**(c)** Calculate the bearing of $C$ from $A$.

..... ° **[3]**

**5** Three towns are located at points $P$, $Q$, and $R$.
Town $Q$ is 24 km from $P$ on a bearing of 042°.
Town $R$ is 35 km from $P$ on a bearing of 118°.



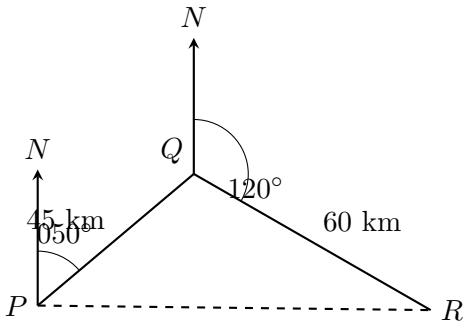

**(a)** Calculate the distance $QR$.

$QR =$ ..... km **[3]**

**(b)** Calculate the bearing of $R$ from $Q$.

..... ° **[4]**

**6** A ship sails 45 km from Port $P$ on a bearing of 050° to Point $Q$.
It then sails 60 km on a bearing of 120° to Point $R$.




**(a)** Calculate the distance $PR$.

..... km **[4]**

**(b)** Calculate the bearing of $R$ from $P$.

..... ° **[3]**

**(c)** The ship sails directly back from $R$ to $P$.
Calculate the bearing of $P$ from $R$.

..... ° **[2]**